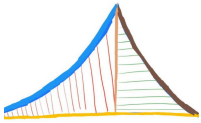
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

License: https://creativecommons.org/licenses/by-sa/3.0/
and
https://www.gnu.org/licenses/fdl-1.3.en.html
First manual appeared in January 2018
First edition published on July 10, 2024
In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

Contents

1 GATE CS2 2026 4
1.1 Formulae 4
1.2 Problems 8

1 GATE CS2 2026

1.1 Formulae

1.1.1 The Z-transform of the PMF of an integer-valued random variable X is defined as

$$M_X(z) = \sum_{k=-\infty}^{\infty} p_X(k)\, z^{-k} = E\left[z^{-X}\right] \tag{1.1.1.1}$$

1.1.2 If $X_1, \ldots, X_n$ are independent and $Y = X_1 + \cdots + X_n$, then

$$M_Y(z) = \prod_{i=1}^{n} M_{X_i}(z) \tag{1.1.2.1}$$

1.1.3 For a Bernoulli random variable X with $\Pr(X = 1) = p$ and $\Pr(X = 0) = q = 1 - p$,

$$M_X(z) = q + p\, z^{-1} \tag{1.1.3.1}$$

1.1.4 If Y is the sum of n independent Bernoulli random variables, each with parameter p, then

$$p_Y(k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \ldots, n \tag{1.1.4.1}$$

1.1.5 The joint Z-transform of two integer-valued random variables X and Y is

$$M_{X,Y}(z_1, z_2) = E\left[z_1^{-X} z_2^{-Y}\right] = \sum_j \sum_k \Pr(X = j, Y = k)\; z_1^{-j} z_2^{-k} \tag{1.1.5.1}$$

so $\Pr(X = j, Y = k)$ is the coefficient of $z_1^{-j} z_2^{-k}$. If $(X, Y) = \sum_i (X_i, Y_i)$ with the pairs (X_i, Y_i) independent, then

$$M_{X,Y}(z_1, z_2) = \prod_i M_{X_i,Y_i}(z_1, z_2) \tag{1.1.5.2}$$

1.1.6 For a fair die X, uniform on $\{1, \ldots, 6\}$,

$$M_X(z) = \frac{1}{6} \sum_{k=1}^{6} z^{-k} \tag{1.1.6.1}$$

If $S = X_1 + X_2$ for two independent fair dice,

$$p_S(k) = \frac{6 - |k - 7|}{36}, \quad k = 2, \ldots, 12 \tag{1.1.6.2}$$

1.1.7 A random variable X is Gaussian (normal), $X \sim \mathcal{N}\left(\mu, \sigma^2\right)$, if its PDF is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in (-\infty, \infty) \tag{1.1.7.1}$$

1.1.8 Other standard distributions:

a) An exponential random variable with rate $\lambda > 0$ is continuous, takes only non-

negative values, and has PDF

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \tag{1.1.8.1}$$

b) A Poisson random variable X with parameter $\lambda > 0$ is discrete. It counts the number of events in a fixed interval when events occur independently at an average rate λ, so it takes only the values $0, 1, 2, \ldots$, with PMF

$$\Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots \tag{1.1.8.2}$$

Being discrete, it has a PMF, not a PDF over the real line.

c) A uniform random variable on $[a, b]$ is continuous with PDF

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \tag{1.1.8.3}$$

which is constant on $[a, b]$ and zero outside it.

1.1.9 Conditional probability:

$$\Pr(A \mid B) = \frac{\Pr(AB)}{\Pr(B)} \tag{1.1.9.1}$$

1.1.10 If g is an odd function, $g(-x) = -g(x)$, then

$$\int_{-a}^{a} g(x)\, dx = 0 \tag{1.1.10.1}$$

1.1.11 For $m \neq -1$,

$$\int x^m\, dx = \frac{x^{m+1}}{m+1} + C \tag{1.1.11.1}$$

1.1.12 The sum of the first n natural numbers is

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2} \tag{1.1.12.1}$$

1.1.13 Asymptotic notation. For positive functions $f(n)$ and $g(n)$,

$$f(n) = o(g(n)) \iff \lim_{n\to\infty} \frac{f(n)}{g(n)} = 0 \tag{1.1.13.1}$$

$$f(n) = O(g(n)) \iff f(n) \leq c\, g(n) \text{ for large } n \tag{1.1.13.2}$$

$$f(n) = \Theta(g(n)) \iff c_1 g(n) \leq f(n) \leq c_2 g(n) \text{ for large } n \tag{1.1.13.3}$$

for some constants $c, c_1, c_2 > 0$. In words, o (little-o) means f grows *strictly slower* than g: however large a multiple of f is taken, g eventually overtakes it. O (big-O) means f grows *no faster* than g, possibly at the same rate. Θ means f and g grow at the *same* rate, up to constant factors. So o is the strict version of O, just as $<$ is the strict version of $\leq$.

1.1.14 For any constants $\epsilon > 0$ and $0 \le a < b$,

$$\log(n) = o\left(n^{\epsilon}\right), \qquad n^a = o\left(n^b\right) \tag{1.1.14.1}$$

1.1.15

$$\log(n!) = \Theta\left(n \log(n)\right) \tag{1.1.15.1}$$

1.1.16 Master theorem. For $T(n) = a\,T\left(\frac{n}{b}\right) + f(n)$ with $a \ge 1$, $b > 1$,

$$T(n) = \begin{cases} \Theta\left(n^{\log_b a}\right) & f(n) = O\left(n^{\log_b a - \epsilon}\right) \text{ for some } \epsilon > 0 \\ \Theta\left(n^{\log_b a} \log(n)\right) & f(n) = \Theta\left(n^{\log_b a}\right) \end{cases} \tag{1.1.16.1}$$

1.1.17 The (one-sided) Z-transform of a sequence $x[k]$, $k \ge 0$, is

$$X(z) = \sum_{k=0}^{\infty} x[k]\, z^{-k} \tag{1.1.17.1}$$

If $x[-1] = 0$, a delay by one step multiplies the transform by z^{-1}:

$$x[k-1] \overset{\mathcal{Z}}{\longleftrightarrow} z^{-1} X(z) \tag{1.1.17.2}$$

1.1.18 For a constant a and $|z| > |a|$,

$$a^k \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1 - az^{-1}} \tag{1.1.18.1}$$

$$(k+1)\,a^k \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1 - az^{-1}\right)^2} \tag{1.1.18.2}$$

$$\frac{(k+1)(k+2)}{2} \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1 - z^{-1}\right)^3} \tag{1.1.18.3}$$

1.1.19 A relation R on a set A is

$$\text{reflexive if } (x,x) \in R \quad \forall x \in A \tag{1.1.19.1}$$

$$\text{symmetric if } (x,y) \in R \implies (y,x) \in R \tag{1.1.19.2}$$

$$\text{transitive if } (x,y),(y,z) \in R \implies (x,z) \in R \tag{1.1.19.3}$$

$$\text{antisymmetric if } (x,y),(y,x) \in R \implies x = y \tag{1.1.19.4}$$

1.1.20 The characteristic equation for a matrix $\mathbf{A}$ is

$$f(\lambda) = \left|\mathbf{A} - \lambda \mathbf{I}\right| = 0 \tag{1.1.20.1}$$

1.1.21 Cayley-Hamilton theorem:

$$f(\lambda) = f(\mathbf{A}) = \mathbf{0} \tag{1.1.21.1}$$

1.1.22 Solving $\mathbf{Ax} = \mathbf{c}$ by row reduction of the augmented matrix $\begin{pmatrix}\mathbf{A} & \mathbf{c}\end{pmatrix}$. Elementary row operations, such as

$$R_2 \leftarrow \alpha R_2 + \beta R_1, \quad \alpha \ne 0 \tag{1.1.22.1}$$

do not change the solutions. After reduction:

a) a row $\begin{pmatrix} 0 & \dots & 0 & | & 0 \end{pmatrix}$ leaves a variable free, giving infinitely many solutions;

b) a row $\begin{pmatrix} 0 & \dots & 0 & | & d \end{pmatrix}$ with $d \neq 0$ means there is no solution.

1.1.23 For an $n \times n$ matrix $\mathbf{A}$ and a scalar c,

$$\det(c\mathbf{A}) = c^n \det(\mathbf{A}) \tag{1.1.23.1}$$

1.1.24 In n-bit two's complement, the bits $b_{n-1} \dots b_1 b_0$ represent

$$x = -b_{n-1}2^{n-1} + \sum_{i=0}^{n-2} b_i 2^i \tag{1.1.24.1}$$

and the representable range is

$$-2^{n-1} \leq x \leq 2^{n-1} - 1 \tag{1.1.24.2}$$

An arithmetic operation overflows if its true result lies outside (1.1.24.2).

1.1.25 IEEE 754 single precision. A 32-bit word with sign bit s, 8-bit exponent E and 23-bit fraction f represents (for $0 < E < 255$)

$$x = (-1)^s \times (1.f)_2 \times 2^{E-127} \tag{1.1.25.1}$$

1.1.26 Cyclic Redundancy Check. For data bits $D(x)$ and a generator $G(x)$ of degree r, the check bits appended to the data are

$$R(x) = x^r D(x) \bmod G(x) \tag{1.1.26.1}$$

computed with modulo-2 (XOR) arithmetic. In bits: append r zeros to the data and divide by the generator pattern.

1.1.27 Boolean algebra:

$$X + X' = 1 \tag{1.1.27.1}$$

$$XY + X'Y = Y \tag{1.1.27.2}$$

1.1.28 An active-high $2:4$ decoder with inputs A (MSB) and B gives

$$D_0 = A'B', \quad D_1 = A'B, \quad D_2 = AB', \quad D_3 = AB \tag{1.1.28.1}$$

A 4-to-1 multiplexer with select lines $S_1 S_0$ and inputs $I_0, \dots, I_3$ gives

$$Y = S_1' S_0' I_0 + S_1' S_0 I_1 + S_1 S_0' I_2 + S_1 S_0 I_3 \tag{1.1.28.2}$$

1.1.29 The Jaccard coefficient of two sets A, B is

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{1.1.29.1}$$

A spanning tree of a connected graph with n vertices has $n - 1$ edges. The complete graph K_n has $\frac{n(n-1)}{2}$ edges.

1.1.30 Hashing with chaining. A key k is placed in the chain of slot

$$h(k) = k \bmod m \tag{1.1.30.1}$$

for a table of m slots. A chain's length is the number of keys that hash to its slot.

1.1.31 Merging sorted arrays L and R outputs L followed by R if and only if

$$\max(L) \le \min(R) \tag{1.1.31.1}$$

1.1.32 A stack is last-in first-out: elements are popped in the reverse of the order in which they were pushed. A queue is first-in first-out: elements are dequeued in the order in which they were enqueued.

1.1.33 Binary semaphore S:

$$\text{Wait}(S) : \text{block until } S = 1, \text{ then set } S = 0 \tag{1.1.33.1}$$

$$\text{Signal}(S) : \text{set } S = 1 \text{ (waking one waiting process)} \tag{1.1.33.2}$$

A binary semaphore initialised to 1, with Wait before and Signal after a block of code, lets only one process run that block at a time (mutual exclusion).

1.1.34 C scope rules. A declaration in an inner block hides a declaration of the same name in an outer scope. The same name cannot be declared twice in the same scope. For any variable `x`,

```
*(&x) = x
```
(1.1.34.1)

1.1.35 The second digit after the decimal point of $r \in (0, 1)$ is

$$d_2(r) = \lfloor 100r \rfloor \bmod 10 \tag{1.1.35.1}$$

so d_2 is constant on each interval $\left[\frac{k}{100}, \frac{k+1}{100}\right)$.

1.2 Problems

1.2.1 A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day? (CS2 2026)

i) $\frac{1}{4}$ ii) $\frac{3}{4}$ iii) $\frac{2}{3}$ iv) $\frac{3}{8}$

Solution:

Variable	Description	Value
X_i	1 if day i is cloudy, 0 if sunny	$\{0, 1\}$
p	$\Pr(X_i = 1)$	0.5
q	$\Pr(X_i = 0) = 1 - p$	0.5
Y	$X_1 + X_2 + X_3 + X_4$	$\{0, \ldots, 4\}$

TABLE 1.2.1: Variables

From (1.1.3.1), the Z-transform of each day is

$$M_{X_i}(z) = q + p\,z^{-1} = \frac{1}{2} + \frac{1}{2}z^{-1} \tag{1.2.1.1}$$

upon substituting from Table 1.2.1. Since the days are independent, from (1.1.2.1) and (1.2.1.1),

$$M_Y(z) = \prod_{i=1}^{4} M_{X_i}(z) = \left(\frac{1}{2} + \frac{1}{2}z^{-1}\right)^4 = \frac{1}{16}\left(1 + z^{-1}\right)^4 \tag{1.2.1.2}$$

$$= \frac{1}{16}\left(1 + 4z^{-1} + 6z^{-2} + 4z^{-3} + z^{-4}\right) \tag{1.2.1.3}$$

$$= \frac{1}{16} + \frac{4}{16}z^{-1} + \frac{6}{16}z^{-2} + \frac{4}{16}z^{-3} + \frac{1}{16}z^{-4} \tag{1.2.1.4}$$

Comparing (1.2.1.4) with (1.1.1.1), $p_Y(k)$ is the coefficient of z^{-k}, so

$$p_Y(3) = \frac{4}{16} = \frac{1}{4} \tag{1.2.1.5}$$

This agrees with (1.1.4.1): $\binom{4}{3}(0.5)^3(0.5)^1 = \frac{1}{4}$.
The following code verifies the solution and generates Fig. 1.2.1.1.

codes/gate_cs2_q3.py

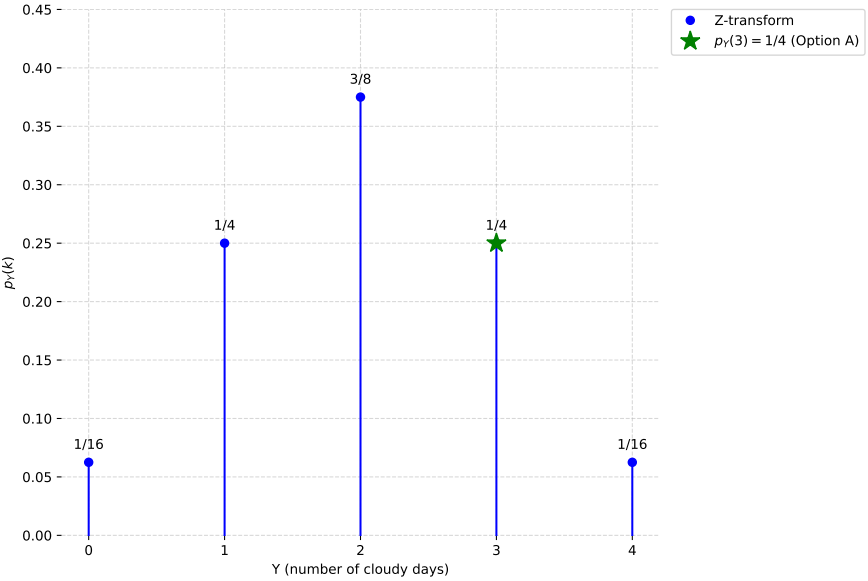


Fig. 1.2.1.1: PMF of the number of cloudy days

1.2.2 The values of Stock A and Stock B on a particular day are Rs. 50 and Rs. 80, respectively. An investor invests Rs. 100 in Stock A and Rs. 80 in Stock B. He sells all the stocks the next day when the value of Stock A is Rs. 55 and Stock B is Rs. 70. The profit made by the investor is Rs. _____ (CS2 2026)

i) 0 ii) 5 iii) 10 iv) 20

Solution: The number of shares bought is

$$\text{Stock A} : \frac{100}{50} = 2, \qquad \text{Stock B} : \frac{80}{80} = 1 \tag{1.2.2.1}$$

and their value the next day is

$$2 \times 55 + 1 \times 70 = 180 \tag{1.2.2.2}$$

so the profit is $180 - (100 + 80) = 0$.
The following code verifies the solution.

codes/stock.c

1.2.3 An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the sum of the two recorded numbers is a prime number is _____ (CS2 2026)

i) $\frac{3}{36}$ ii) $\frac{13}{36}$ iii) $\frac{15}{36}$ iv) $\frac{19}{36}$

Solution: Let X_1, X_2 be the two numbers and $S = X_1 + X_2$. From (1.1.6.1), the Z-transform of each roll is

$$M_{X_i}(z) = \frac{1}{6}\left(z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6}\right) \tag{1.2.3.1}$$

The two rolls are independent, so from (1.1.2.1) and (1.2.3.1),

$$M_S(z) = M_{X_1}(z)\, M_{X_2}(z) = \frac{1}{36}\left(z^{-1} + z^{-2} + \cdots + z^{-6}\right)^2 \tag{1.2.3.2}$$

$$= \frac{1}{36}(z^{-2} + 2z^{-3} + 3z^{-4} + 4z^{-5} + 5z^{-6} + 6z^{-7} + 5z^{-8} + 4z^{-9} + 3z^{-10} + 2z^{-11} + z^{-12}) \tag{1.2.3.3}$$

Comparing (1.2.3.3) with (1.1.1.1), $p_S(k)$ is the coefficient of z^{-k}, which agrees with (1.1.6.2).

k	2	3	4	5	6	7	8	9	10	11	12
$36\,p_S(k)$	1	2	3	4	5	6	5	4	3	2	1

TABLE 1.2.3: Coefficients of $M_S(z)$

The primes in $\{2, \ldots, 12\}$ are $2, 3, 5, 7, 11$. From Table 1.2.3,

$$\Pr(S \text{ prime}) = p_S(2) + p_S(3) + p_S(5) + p_S(7) + p_S(11) \tag{1.2.3.4}$$

$$= \frac{1 + 2 + 4 + 6 + 2}{36} = \frac{15}{36} \tag{1.2.3.5}$$

The following code verifies the solution by simulation and generates Fig. 1.2.3.1.

codes/dice_sum.py

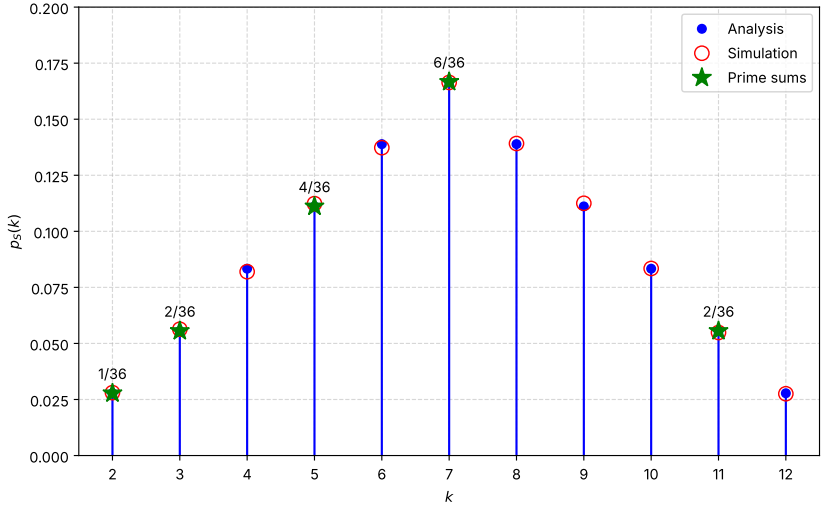


Fig. 1.2.3.1: PMF of the sum of two dice, with the prime sums marked

1.2.4 The probability density function $f(x)$ of a random variable X which takes real values is

$$f(x) = \frac{1}{3\sqrt{2\pi}} \exp\left(-\frac{x^2}{18}\right), \quad x \in (-\infty, +\infty) \tag{1.2.4.1}$$

Which one of the following statements is correct about the random variable X? (CS2 2026)

i) X is an exponential random variable
ii) X is a normal random variable
iii) X is a Poisson random variable
iv) X is a uniform random variable

Solution: X is a normal random variable. Comparing (1.2.4.1) with (1.1.7.1),

$$\mu = 0, \quad 2\sigma^2 = 18 \implies \sigma = 3 \tag{1.2.4.2}$$

so $X \sim \mathcal{N}(0, 9)$.

The other options are false:

- **i) Exponential.** From (1.1.8.1), an exponential PDF is 0 for all $x < 0$. But (1.2.4.1) is positive for every real x, for example

$$f(-3) = \frac{1}{3\sqrt{2\pi}} e^{-1/2} > 0 \tag{1.2.4.3}$$

It is also symmetric, $f(-x) = f(x)$, while an exponential PDF is decreasing from $x = 0$.

- **iii) Poisson.** From (1.1.8.2), a Poisson random variable is discrete: it takes only the values $0, 1, 2, \ldots$ and is described by a PMF. But X takes all real values in $(-\infty, \infty)$, including negative and non-integer values, and is described by a PDF.
- **iv) Uniform.** From (1.1.8.3), a uniform PDF is constant on $[a, b]$ and zero outside it. But (1.2.4.1) changes with x, for example

$$f(0) = \frac{1}{3\sqrt{2\pi}} \neq f(3) = \frac{1}{3\sqrt{2\pi}} e^{-1/2} \tag{1.2.4.4}$$

  and it is never zero.

The following code compares the numerical PDF of $\mathcal{N}(0, 9)$ samples with (1.2.4.1) in Fig. 1.2.4.1.

codes/gauss.py

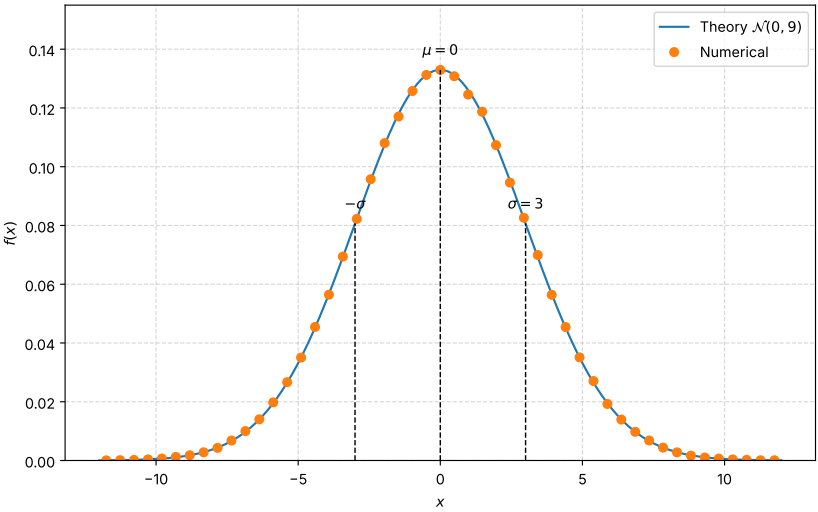


Fig. 1.2.4.1: Numerical PDF vs $\mathcal{N}(0, 9)$

1.2.5 Consider the following three ANSI-C programs, P1, P2, and P3. (CS2 2026)

P1

```
#include <stdio.h>
int a=5;
int main(){
  int a=7;
  return(0);
}
```

P2

```
#include <stdio.h>
int main(){
  int a=5;
  int a=7;
  return(0);
}
```

P3

```
#include <stdio.h>
int main(){
  int a=5;
  float a=7;
  return(0);
}
```

Which one of the following statements is true?

i) Only P1 will compile without any error

ii) Only P2 will compile without any error

iii) Only P3 will compile without any error
iv) All three programs P1, P2, and P3 will compile without any error

Solution: From 1.1.34, a scope is the region of code in which a name is visible. Everything outside all functions is the *file scope*, and each pair of braces { } starts a new *block scope*.

- **P1 compiles.** `int a=5;` is outside `main`, in file scope, while `int a=7;` is inside the braces of `main`, in block scope. These are two different variables in two different scopes. Inside `main` the local `a` hides the global one, which is allowed.
- **P2 does not compile: redefinition.** Both `int a=5;` and `int a=7;` are inside the same braces of `main`, so they are in the same block scope. The compiler would have to create two variables with the same name in the same place, and then could not tell which one a later use of `a` refers to. C forbids this. Note that `int a=7;` is a new declaration, not an assignment. Writing `a=7;` instead would compile and simply change the value of `a`. The compiler reports

```
p2.c:5:7: error: redefinition of 'a'
```

- **P3 does not compile: conflicting types.** This is the same mistake, declaring `a` twice in the same block scope, but now with different types: first `int`, then `float`. A variable has exactly one type for its whole lifetime, so `a` cannot be both an `int` and a `float`. Changing the type does not make it a different variable. The compiler reports

```
p3.c:5:9: error: conflicting types for 'a'; have 'float'
```

P1 shows that the same name may be reused in a *different* scope. P2 and P3 fail because they reuse it in the *same* scope, whatever the type.
Compiling the three programs confirms the solution: only P1 compiles.

```
codes/p1.c
codes/p2.c
codes/p3.c
```

1.2.6 Consider the following functions, where n is a positive integer.

$$n^{1/3}, \quad \log(n), \quad \log(n!), \quad 2^{\log(n)}$$

Which one of the following options lists the functions in increasing order of asymptotic growth rate?
Note: Assume the base of log to be 2. (CS2 2026)

i) $\log(n)$, $n^{1/3}$, $2^{\log(n)}$, $\log(n!)$
ii) $n^{1/3}$, $\log(n)$, $\log(n!)$, $2^{\log(n)}$
iii) $\log(n)$, $n^{1/3}$, $\log(n!)$, $2^{\log(n)}$
iv) $2^{\log(n)}$, $n^{1/3}$, $\log(n)$, $\log(n!)$

Solution: With base 2, $2^{\log(n)} = n$. From (1.1.15.1), $\log(n!) = \Theta(n \log(n))$, so $\log(n!)$ grows at the same rate as $n \log(n)$ (see (1.1.13.3)). It remains to order $\log(n)$, $n^{1/3}$,

n and $n\log(n)$. Here o is little-o from (1.1.13.1): $f = o(g)$ means $\frac{f}{g} \to 0$, that is, f grows strictly slower than g. From (1.1.14.1),

$$\log(n) = o\left(n^{1/3}\right), \quad n^{1/3} = o(n), \quad n = o\left(n\log(n)\right) \tag{1.2.6.1}$$

Each ratio goes to 0:

Statement	Ratio	Why it goes to 0
$\log(n) = o\left(n^{1/3}\right)$	$\frac{\log(n)}{n^{1/3}}$	a power of n beats a logarithm; at $n = 10^8$: $\log(n) \approx 27$, $n^{1/3} \approx 464$
$n^{1/3} = o(n)$	$\frac{n^{1/3}}{n} = \frac{1}{n^{2/3}}$	$n^{2/3} \to \infty$
$n = o\left(n\log(n)\right)$	$\frac{n}{n\log(n)} = \frac{1}{\log(n)}$	$\log(n) \to \infty$

TABLE 1.2.6: Ratios in (1.2.6.1)

Chaining (1.2.6.1) gives $\log(n) < n^{1/3} < n < n\log(n)$ in growth rate. Since $2^{\log(n)} = n$ and $\log(n!) = \Theta\left(n\log(n)\right)$, the order is $\log(n)$, $n^{1/3}$, $2^{\log(n)}$, $\log(n!)$.
The following code plots the four functions in Fig. 1.2.6.1. Note that $n^{1/3}$ overtakes $\log(n)$ only after $n \approx 10^3$.

codes/growth.py

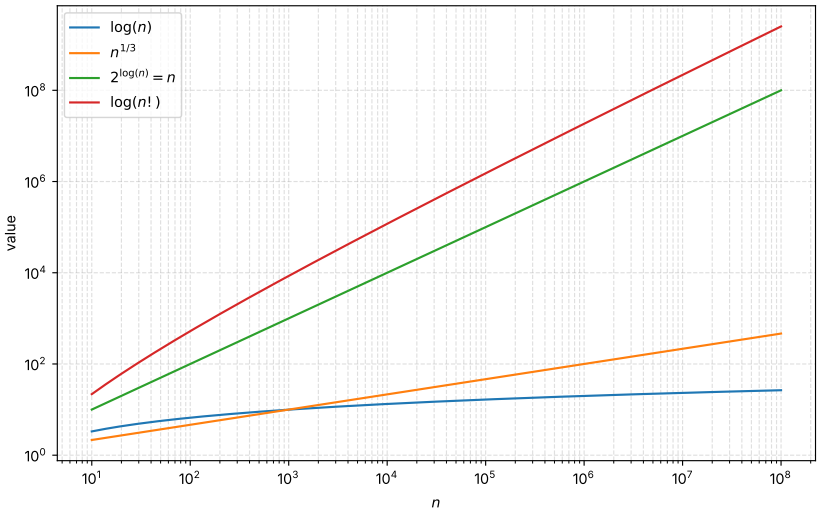


Fig. 1.2.6.1: Growth of the four functions

1.2.7 Which of the following can be recurrence relation(s) corresponding to an algorithm with time complexity $\Theta(n)$? (CS2 2026)

i) $T(n) = T(n-1) + 1,\ T(1) = 1$
ii) $T(n) = 2T\left(\frac{n}{2}\right) + 1,\ T(1) = 1$
iii) $T(n) = 2T\left(\frac{n}{2}\right) + n,\ T(1) = 1$
iv) $T(n) = T(n-1) + n,\ T(1) = 1$

Solution: Each recurrence is turned into a sequence $x[k]$, $k \geq 0$, and solved with the Z-transform. In each case setting $x[-1] = 0$ reproduces the initial condition, so (1.1.17.2) applies.

i) Let $x[k] = T(k+1)$. Then

$$x[k] = x[k-1] + 1, \quad x[0] = 1 \tag{1.2.7.1}$$

Taking the Z-transform using (1.1.17.2) and (1.1.18.1) with $a = 1$,

$$X(z) = z^{-1}X(z) + \frac{1}{1-z^{-1}} \implies X(z) = \frac{1}{(1-z^{-1})^2} \tag{1.2.7.2}$$

From (1.1.18.2) with $a = 1$, $x[k] = k + 1$, so

$$T(n) = x[n-1] = n = \Theta(n) \tag{1.2.7.3}$$

ii) Let $n = 2^k$ and $x[k] = T\left(2^k\right)$. Since $T\left(\frac{n}{2}\right) = x[k-1]$,

$$x[k] = 2x[k-1] + 1, \quad x[0] = 1 \tag{1.2.7.4}$$

Taking the Z-transform,

$$X(z) = 2z^{-1}X(z) + \frac{1}{1-z^{-1}} \implies X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})} \tag{1.2.7.5}$$

By partial fractions,

$$X(z) = \frac{2}{1-2z^{-1}} - \frac{1}{1-z^{-1}} \tag{1.2.7.6}$$

so from (1.1.18.1), $x[k] = 2 \cdot 2^k - 1$, and

$$T(n) = 2n - 1 = \Theta(n) \tag{1.2.7.7}$$

iii) With $n = 2^k$ and $x[k] = T\left(2^k\right)$,

$$x[k] = 2x[k-1] + 2^k, \quad x[0] = 1 \tag{1.2.7.8}$$

Taking the Z-transform using (1.1.18.1) with $a = 2$,

$$X(z) = 2z^{-1}X(z) + \frac{1}{1-2z^{-1}} \implies X(z) = \frac{1}{(1-2z^{-1})^2} \tag{1.2.7.9}$$

From (1.1.18.2) with $a = 2$, $x[k] = (k+1)\,2^k$. Since $k = \log(n)$,

$$T(n) = n\,(\log(n) + 1) = \Theta\left(n \log(n)\right) \tag{1.2.7.10}$$

iv) Let $x[k] = T(k+1)$. Then

$$x[k] = x[k-1] + (k+1), \quad x[0] = 1 \tag{1.2.7.11}$$

Taking the Z-transform using (1.1.18.2) with $a = 1$,

$$X(z) = z^{-1}X(z) + \frac{1}{(1-z^{-1})^2} \implies X(z) = \frac{1}{(1-z^{-1})^3} \tag{1.2.7.12}$$

From (1.1.18.3), $x[k] = \frac{(k+1)(k+2)}{2}$, so

$$T(n) = x[n-1] = \frac{n(n+1)}{2} = \Theta\left(n^2\right) \tag{1.2.7.13}$$

which agrees with (1.1.12.1).

Only i) and ii) are $\Theta(n)$, in agreement with the master theorem (1.1.16.1).
The following code checks each closed form above against the recurrence and plots $T(n)/n$ in Fig. 1.2.7.1. It stays bounded only for i) and ii).

`codes/recurrence.py`

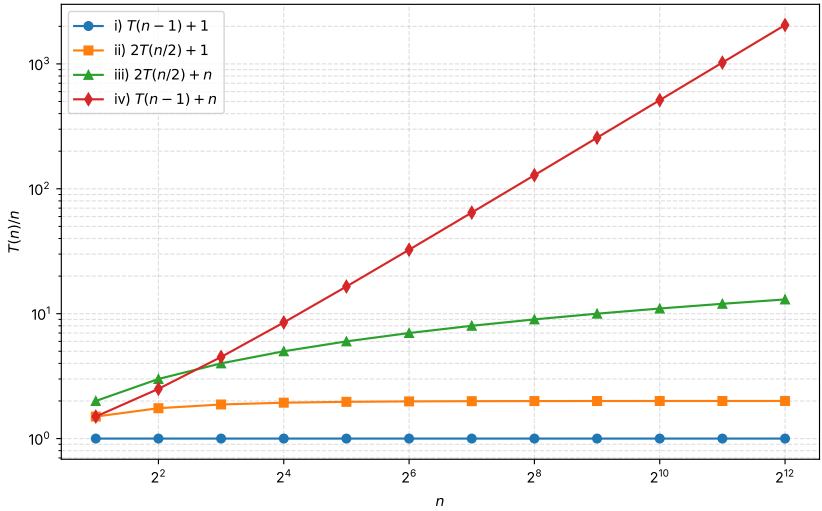


Fig. 1.2.7.1: $T(n)/n$ for the four recurrences

1.2.8 Let R be a binary relation on the set $\{1, 2, \ldots, 10\}$, where $(x, y) \in R$ if the product of x and y is square of an integer. Which of the following properties is/are satisfied by R? (CS2 2026)

i) Reflexive ii) Symmetric iii) Transitive iv) Antisymmetric

Solution: Checking each property against (1.1.19.1)–(1.1.19.4):

- Reflexive: $x \cdot x = x^2$.
- Symmetric: $xy = yx$.

- Transitive: if $xy = a^2$ and $yz = b^2$, then

$$xz \cdot y^2 = a^2 b^2 \implies xz = \left(\frac{ab}{y}\right)^2 \tag{1.2.8.1}$$

An integer that is the square of a rational is the square of an integer, so $(x, z) \in R$.

- Not antisymmetric: $(1, 4), (4, 1) \in R$ since $1 \times 4 = 2^2$, but $1 \neq 4$.

The following code checks all four properties and draws R in Fig. 1.2.8.1.

codes/relation.py

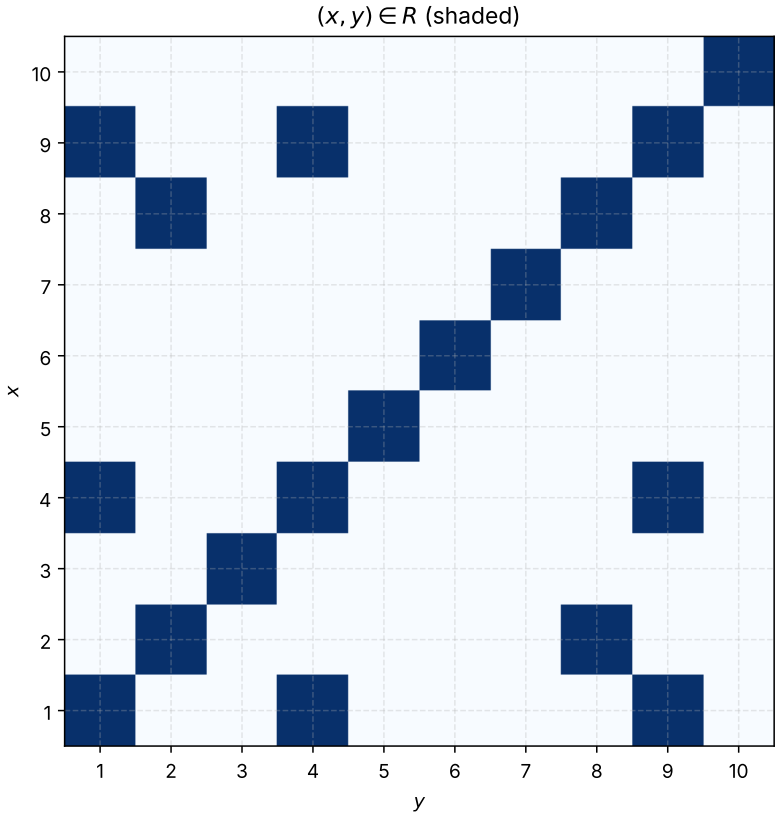


Fig. 1.2.8.1: The relation R on $\{1, \ldots, 10\}$

1.2.9 For a real number a, let

$$I(a) = \int_{-1}^{1} \left(3x^2 - ax + 1\right) dx. \tag{1.2.9.1}$$

Which of the following statements is/are true? (CS2 2026)

i) The value of $I(a)$ is independent of the value of a

ii) The value of $I(a)$ can vary with the value of a
iii) There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a positive real number
iv) There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a negative real number

Solution: Since ax is odd, from (1.1.10.1) and (1.1.11.1),

$$I(a) = \left[x^3 + x\right]_{-1}^{1} - a \cdot 0 = 2 - (-2) = 4 \tag{1.2.9.2}$$

for every a. So i) and iii) are true and ii) and iv) are false.
The following code evaluates $I(a)$ numerically in Fig. 1.2.9.1.

codes/integral.py

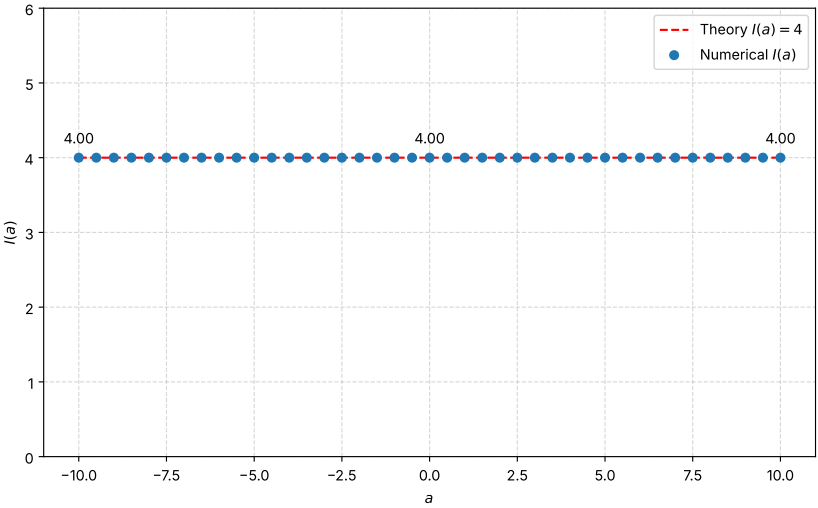


Fig. 1.2.9.1: $I(a)$ for $-10 \le a \le 10$

1.2.10 In a system, numbers are represented using 4-bit two's complement form. Consider four numbers $N_1 = 1011$, $N_2 = 1101$, $N_3 = 1010$ and $N_4 = 1001$ in the system. Which of the following operations will result in arithmetic overflow? (CS2 2026)

i) $N_1 + N_2$ ii) $N_2 + N_3$ iii) $N_3 - N_4$ iv) $N_1 + N_4$

Solution: From (1.1.24.1),

$$N_1 = -5, \quad N_2 = -3, \quad N_3 = -6, \quad N_4 = -7 \tag{1.2.10.1}$$

and from (1.1.24.2) the range is $-8 \le x \le 7$.

Option	Operation	Result	Overflow?
i)	$-5+(-3)$	-8	No
ii)	$-3+(-6)$	-9	Yes
iii)	$-6-(-7)$	1	No
iv)	$-5+(-7)$	-12	Yes

TABLE 1.2.10: Operations

The following code verifies the solution.

codes/twos.c

1.2.11 The keys 5, 28, 19, 15, 26, 33, 12, 17, 10 are inserted into a hash table using the hash function $h(k) = k \bmod 9$. The collisions are resolved by chaining. After all the keys are inserted, the length of the longest chain is _____. (*answer in integer*) (CS2 2026)

Solution: From (1.1.30.1) with $m = 9$:

k	5	28	19	15	26	33	12	17	10
$h(k)$	5	1	1	6	8	6	3	8	1

TABLE 1.2.11: Hash values

Slot 1 holds 28, 19, 10, so the longest chain has length 3.
The following code builds the chains and draws them in Fig. 1.2.11.1.

codes/hash.py

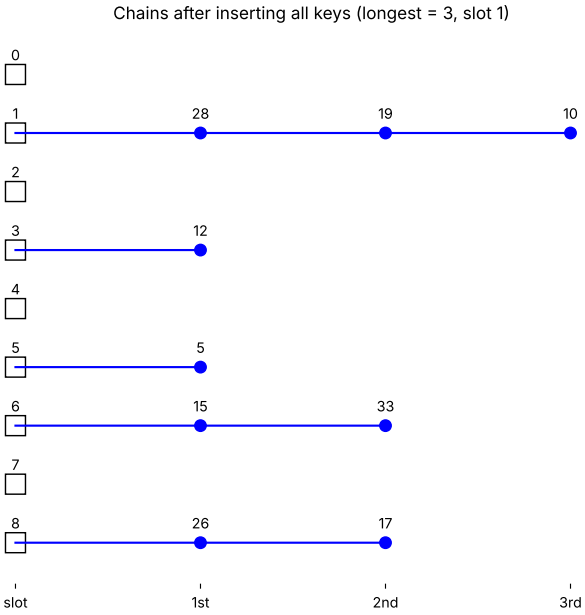


Fig. 1.2.11.1: Chains in the hash table

1.2.12 Consider the system of linear equations given below.

$$ax + y = b \tag{1.2.12.1}$$

$$16x + ay = 24 \tag{1.2.12.2}$$

Suppose the values of a and b are chosen such that the system of linear equations produce multiple solutions. Then the product of a and b is _____. (*answer in integer*) (CS2 2026)

Solution: This is solved using row reduction 1.1.22, the characteristic equation (1.1.20.1) and the Cayley-Hamilton theorem (1.1.21.1). The system is $\mathbf{Ax} = \mathbf{c}$ with augmented matrix

$$\begin{pmatrix}\mathbf{A} & \mathbf{c}\end{pmatrix} = \begin{pmatrix} a & 1 & b \\ 16 & a & 24 \end{pmatrix} \tag{1.2.12.3}$$

If $a = 0$, the equations are $y = b$ and $16x = 24$, which have the unique solution $\left(\frac{3}{2}, b\right)$. So $a \neq 0$, and using (1.1.22.1) with $R_2 \leftarrow aR_2 - 16R_1$,

$$\begin{pmatrix} a & 1 & b \\ 16 & a & 24 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow aR_2 - 16R_1} \begin{pmatrix} a & 1 & b \\ 0 & a^2 - 16 & 24a - 16b \end{pmatrix} \tag{1.2.12.4}$$

From 1.1.22, there are infinitely many solutions only if the second row of (1.2.12.4)

is entirely zero:

$$a^2 - 16 = 0 \tag{1.2.12.5}$$

$$24a - 16b = 0 \tag{1.2.12.6}$$

Solving (1.2.12.5) with the characteristic equation. Let

$$\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 16 & 0 \end{pmatrix} \tag{1.2.12.7}$$

From (1.1.20.1), its characteristic equation is

$$f(\lambda) = \begin{vmatrix} -\lambda & 1 \\ 16 & -\lambda \end{vmatrix} = \lambda^2 - 16 = 0 \implies \lambda = \pm 4 \tag{1.2.12.8}$$

and from (1.1.21.1), $\mathbf{N}^2 - 16\mathbf{I} = \mathbf{0}$. Since $a^2 - 16 = f(-a)$, (1.2.12.5) holds exactly when $-a$ is a root of (1.2.12.8), so $a = \pm 4$.
Solving (1.2.12.6). $b = \frac{24a}{16} = \frac{3a}{2}$, so

$$a = 4 \implies b = 6 \tag{1.2.12.9}$$

$$a = -4 \implies b = -6 \tag{1.2.12.10}$$

Checking by row reduction. For $a = 4$, $b = 6$,

$$\begin{pmatrix} 4 & 1 & 6 \\ 16 & 4 & 24 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2 - 4R_1} \begin{pmatrix} 4 & 1 & 6 \\ 0 & 0 & 0 \end{pmatrix} \xleftrightarrow{R_1 \leftarrow \frac{R_1}{4}} \begin{pmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ 0 & 0 & 0 \end{pmatrix} \tag{1.2.12.11}$$

The zero row leaves y free:

$$x = \frac{3}{2} - \frac{1}{4}y \implies \mathbf{x} = \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} + y \begin{pmatrix} -\frac{1}{4} \\ 1 \end{pmatrix} \tag{1.2.12.12}$$

For $a = -4$, $b = -6$,

$$\begin{pmatrix} -4 & 1 & -6 \\ 16 & -4 & 24 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2 + 4R_1} \begin{pmatrix} -4 & 1 & -6 \\ 0 & 0 & 0 \end{pmatrix} \xleftrightarrow{R_1 \leftarrow -\frac{R_1}{4}} \begin{pmatrix} 1 & -\frac{1}{4} & \frac{3}{2} \\ 0 & 0 & 0 \end{pmatrix} \tag{1.2.12.13}$$

so

$$\mathbf{x} = \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} + y \begin{pmatrix} \frac{1}{4} \\ 1 \end{pmatrix} \tag{1.2.12.14}$$

Both cases have infinitely many solutions, and in both

$$ab = 24 \tag{1.2.12.15}$$

The following code draws the solution lines (1.2.12.12) and (1.2.12.14) in Fig. 1.2.12.1.

codes/lines.py

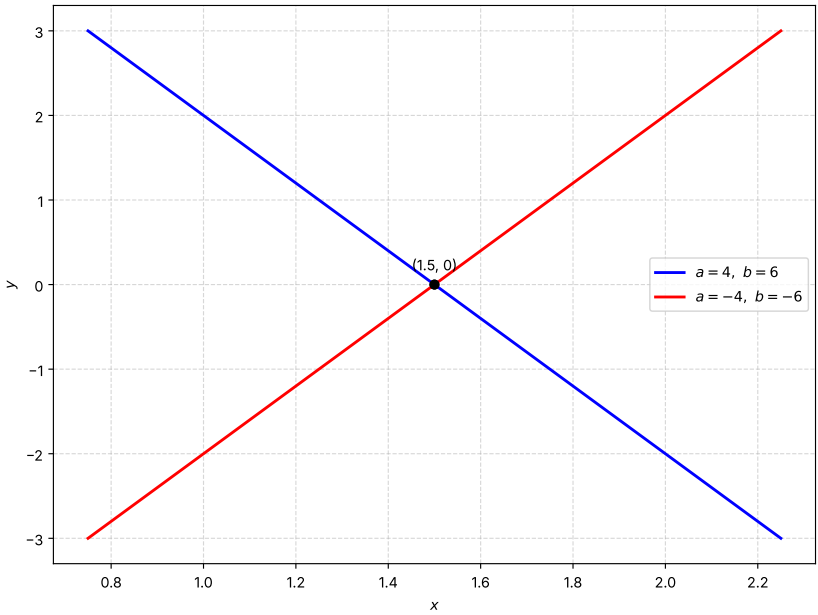


Fig. 1.2.12.1: Solution lines for $(a, b) = (4, 6)$ and $(-4, -6)$

1.2.13 Consider an array $A = [10, 7, 8, 19, 41, 35, 25, 31]$. Suppose the merge sort algorithm is executed on array A to sort it in increasing order. The merge sort algorithm will carry out a total of 7 merge operations.

A merge operation on sorted left array L and sorted right array R is said to be void if the output of the merge operation is the elements of array L followed by the elements of array R.

The number of void merge operations among these 7 merge operations is _____. (*answer in integer*) (CS2 2026)

Solution: Applying (1.1.31.1) to each merge:

L	R	**Void?**
$[10]$	$[7]$	No
$[8]$	$[19]$	Yes
$[41]$	$[35]$	No
$[25]$	$[31]$	Yes
$[7, 10]$	$[8, 19]$	No
$[35, 41]$	$[25, 31]$	No
$[7, 8, 10, 19]$	$[25, 31, 35, 41]$	Yes

TABLE 1.2.13: Merge operations

There are 3 void merges.

The following code verifies the solution.

codes/merge.c

1.2.14 The 32-bit IEEE 754 single precision representation of a number is `0xC2710000`. The number in decimal representation is _____. (*rounded off to two decimal places*) (CS2 2026)

Solution: In binary,

$$\texttt{0xC2710000} = 1 \quad 10000100 \quad 11100010000000000000000 \tag{1.2.14.1}$$

so $s = 1$, $E = 132$ and $(1.f)_2 = 1.1110001_2 = 1.8828125$. From (1.1.25.1),

$$x = -1.8828125 \times 2^{132-127} = -60.25 \tag{1.2.14.2}$$

The following code verifies the solution.

codes/ieee.c

1.2.15 Consider a complete graph K_n with n vertices ($n > 4$). Note that multiple spanning trees can be constructed over K_n. Each of these spanning trees is represented as a set of edges. The Jaccard coefficient between any two sets is defined as the ratio of the size of the intersection of the two sets to the size of the union of the two sets. Which one of the following options gives the lowest possible value for the Jaccard coefficient between any two spanning trees of K_n? (CS2 2026)

i) $\frac{1}{n}$ ii) $\frac{1}{2n-3}$ iii) 0 iv) $\frac{1}{n-1}$

Solution: From (1.1.29.1), $J \geq 0$, with $J = 0$ when the two trees share no edge. K_n has $\frac{n(n-1)}{2} \geq 2(n-1)$ edges, enough for two edge-disjoint spanning trees of $n-1$ edges each, and such trees exist (for example, edge-disjoint Hamiltonian paths). In K_5,

$$T_1 : 0-1-2-3-4, \qquad T_2 : 0-2-4-1-3 \tag{1.2.15.1}$$

share no edge.

The following code enumerates all 125 spanning trees of K_5, finds the minimum Jaccard coefficient, and draws Fig. 1.2.15.1.

codes/spanning.py

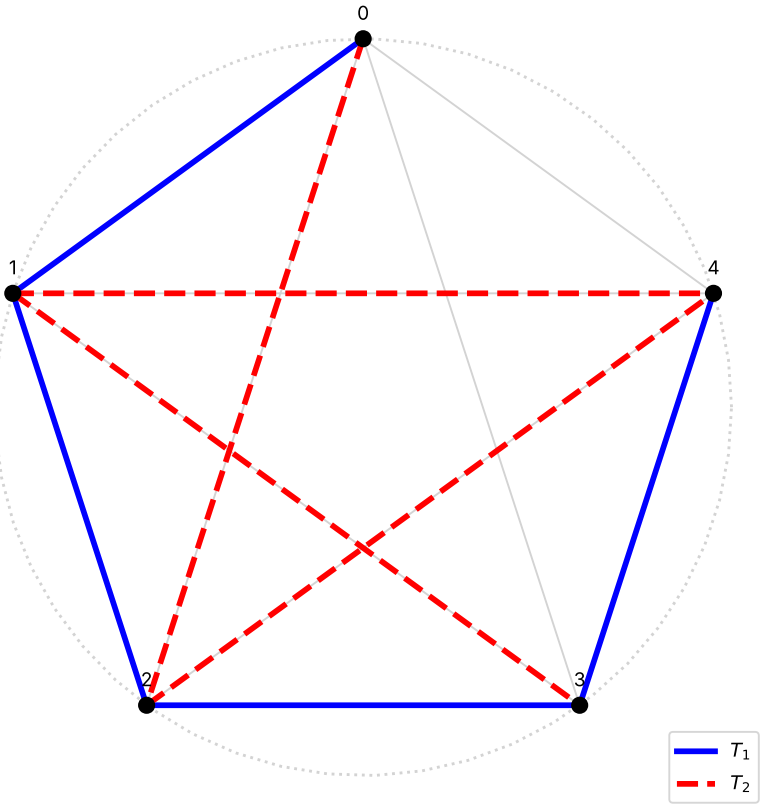


Fig. 1.2.15.1: Edge-disjoint spanning trees T_1, T_2 of K_5

1.2.16 Consider an array A of integers of size n. The indices of A run from 1 to n. An algorithm is to be designed to check whether A satisfies the condition given below.

$$\forall i, j \in \{1, \ldots, n-1\} \text{ such that } i > j, \quad (A[i+1] - A[i]) > (A[j+1] - A[j])$$

Which one of the following gives the worst case time complexity of the fastest algorithm that can be designed for the problem? (CS2 2026)

i) $\Theta(n)$ ii) $\Theta(\log(n))$ iii) $\Theta(n \log(n))$ iv) $\Theta(n^2)$

Solution: Let $d_i = A[i+1] - A[i]$. The condition is

$$d_1 < d_2 < \cdots < d_{n-1} \tag{1.2.16.1}$$

which needs only the consecutive checks $d_i < d_{i+1}$: one pass, $O(n)$. Every element must be read, since changing an unread element could break the condition, so $\Omega(n)$ is needed.
The following code implements the $O(n)$ check.

codes/check.c

1.2.17 Consider a table T, where the elements $T[i][j]$, $0 \le i, j \le n$, represent the cost of the optimal solutions of different subproblems of a problem that is being solved using a dynamic programming algorithm. The recursive formulation to compute the table entries is as follows:

$$T[0][k] = T[k][0] = 1 \qquad k = 0, 1, \ldots, n \qquad (1.2.17.1)$$

$$T[i][j] = 2T[i-1][j] + 3T[i][j-1] \qquad 1 \le i, j \le n \qquad (1.2.17.2)$$

Consider the following two algorithms to compute entries of T. Assume that for both the algorithms, for all $0 \le i, j \le n$, $T[i][j]$ has been initialized to 1.

```
Algorithm B1: For i = 1,2,...,n
                  For j = 1,2,...,n
                    T[i][j] = 2T[i-1][j] + 3T[i][j-1]

Algorithm B2: For s = 2,3,...,2n
                  For i = 1,2,...,n
                    For j = 1,2,...,n
                      If (i + j == s)
                        T[i][j] = 2T[i-1][j] + 3T[i][j-1]
```

Algorithm B_k, $k \in \{1, 2\}$ is said to be correct if and only if it calculates the correct values of $T[i][j]$, for all $0 \le i, j \le n$, (as per the recursive formulation) at the end of the execution of the algorithm B_k. Which one of the following statements is true? (CS2 2026)

i) Both algorithms B_1 and B_2 are correct
ii) Algorithm B_1 is correct, but algorithm B_2 is incorrect
iii) Algorithm B_2 is correct, but algorithm B_1 is incorrect
iv) Both algorithms B_1 and B_2 are incorrect

Solution: In (1.2.17.2), $T[i][j]$ needs $T[i-1][j]$ and $T[i][j-1]$, and the boundary cells are already correct.

- B_1 goes row by row: $T[i-1][j]$ is in the previous row and $T[i][j-1]$ is earlier in the same row. Correct.
- B_2 goes by anti-diagonal $s = i + j$: both dependencies have index sum $s - 1$, so they were computed in the previous pass. Every cell has $2 \le i + j \le 2n$. Correct.

The following code compares both algorithms with the recursive formulation.

codes/dp.c

1.2.18 Consider the following 4-variable Boolean function

$$F(A, B, C, D) = \Sigma m\,(0, 1, 2, 3, 8, 9, 10, 11)$$

Consider A as MSB, D as LSB. Which one of the following options represents the minimal sum of products form for the above function?

Note: + is OR operation, . is AND operation, ′ is NOT operation (CS2 2026)

i) $A' + B' + C' + D'$
ii) B'
iii) $A'.B' + A.B$
iv) A'

Solution: Minterms 0–3 are $00CD$ ($A = 0, B = 0$) and minterms 8–11 are $10CD$ ($A = 1, B = 0$), with C, D taking all values. So

$$F = A'B' + AB' = B' \tag{1.2.18.1}$$

from (1.1.27.2).
The following code checks every option against the truth table.

codes/bool.c

1.2.19 Consider the transmission of data bits 110001011 over a link that uses Cyclic Redundancy Check (CRC) code for error detection. If the generator bit pattern is given to be 1001, which one of the following options shows the remainder bit pattern appended to the data bits before transmission? (CS2 2026)

i) 011 ii) 101 iii) 000 iv) 100

Solution: The generator has degree $r = 3$. From (1.1.26.1), append three zeros and divide by 1001 modulo 2:

```
110001011000
1001
-------------
010101011000
 1001
000111011000
   1001
000011111000
    1001
000001101000
     1001
000000100000
      1001
000000000100
```

The remainder is 100.
The following code verifies the solution.

codes/crc.c

1.2.20 Consider a stack S and a queue Q. Both of them are initially empty and have the capacity to store ten elements each. The elements 1, 2, 3, 4, and 5 arrive one by one, in that order. When an element arrives, it is assigned either to S (pushed on S) or to Q (enqueued to Q). Once all the five elements are stored, the output is generated

in two steps. First, stack S is emptied by popping all elements. Then queue Q is emptied by dequeueing all elements. The output obtained by following this process is 4 3 1 2 5.

Given the output, the objective is to predict whether an element was assigned to S or Q. Which of the following options is/are possible valid assignment(s) of the elements?

Note: In the options, the notation xS denotes that element x was assigned to S and yQ denotes that element y was assigned to Q. (CS2 2026)

i) $1S, 2Q, 3S, 4S, 5Q$
ii) $1Q, 2Q, 3S, 4S, 5Q$
iii) $1Q, 2Q, 3Q, 4S, 5S$
iv) $1S, 2S, 3S, 4Q, 5Q$

Solution: From 1.1.32, the stack's elements come out in decreasing order and the queue's in increasing order. So the output must be a decreasing part (stack) followed by an increasing part (queue).

Stack	Queue	Valid?
4	3 1 2 5	No
4 3	1 2 5	Yes: option ii)
4 3 1	2 5	Yes: option i)
4 3 1 2	5	No

TABLE 1.2.20: Possible splits of the output

Option iii) would output 5 first and option iv) would output 3 2 1 4 5.
The following code tries all 32 assignments and prints the valid ones.

codes/stackqueue.c

1.2.21 Consider three processes P1, P2, and P3 running identical code, as shown in the pseudocode below. A and B are two binary semaphores initialized to 1 and 0, respectively. X is a shared variable initialized to 0. Each line in the pseudocode is executed atomically.

Pseudocode of P1, P2, and P3

```
Wait(A);
Print(*);
X = X+1;
If (X == 2)
   {
      Print($);
      Signal(B);
   }
Signal(A);
Wait(B);
Print(#);
Signal(B);
```

Assume that any of the three processes can start to execute first and context switching can happen between these processes at any arbitrary time and in any arbitrary order. Which of the following patterns is/are possible to be generated as an outcome of the execution of these three processes? (CS2 2026)

i) `**$*###` ii) `**$#*##` iii) `**$###*#` iv) `***$###`

Solution: From (1.1.33.1) and (1.1.33.2), A gives mutual exclusion, so the second process to enter prints `$` right after its `*`, before releasing A. Every output therefore starts with `**$`, which rules out iv). `Signal(B)` after `$`, and after each `#`, lets the waiting processes print their `#` one at a time. The third process can enter at any time after A is released: before any `#` (option i), after one (option ii) or after two (option iii).
The following code explores every interleaving and lists all possible outputs.

```
codes/semaphore.py
```

1.2.22 Consider the digital circuit shown in Fig. 1.2.22.1 with two input lines A and B, two select lines S0 and S1, and an output line Y. The blocks Q and M represent active high 2:4 decoder and 4-to-1 multiplexer, respectively. Out of 16 possible input combinations, the number of combinations that produce Y=1 is _____. (*answer in integer*)
Note: One input combination is an instance of [A B S1 S0]. (CS2 2026)

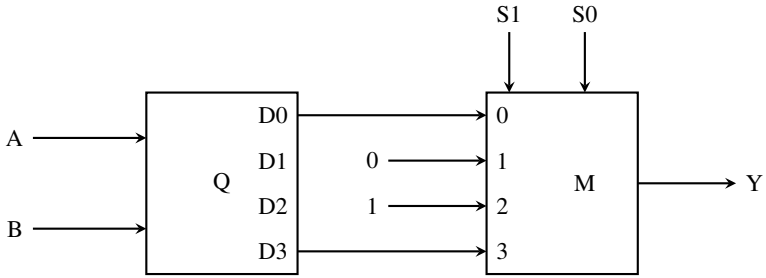


Fig. 1.2.22.1: Decoder and multiplexer circuit

Solution: From (1.1.28.1) and (1.1.28.2), with $I_0 = D_0$, $I_1 = 0$, $I_2 = 1$, $I_3 = D_3$,

$$Y = S_1' S_0' A' B' + S_1 S_0' + S_1 S_0 AB \tag{1.2.22.1}$$

S_1S_0	$Y = 1$ when	**Count**
00	$A = B = 0$	1
01	never	0
10	always	4
11	$A = B = 1$	1

TABLE 1.2.22: Combinations with $Y = 1$

So 6 combinations give $Y = 1$.
The following code prints the truth table and counts $Y = 1$.

codes/circuit.c

1.2.23 Consider the following ANSI-C program.

```c
#include <stdio.h>
int main(){
    int *ptr, a, b, c;
    a=5; b=11; c=20;
    ptr=&a; *ptr=c; ptr=&c;
    a=*(&b); c=*ptr-a;
    printf("%d",c);
    return(0);
}
```

The output of this program is _____. (*answer in integer*)
Note: Assume that the program compiles and runs successfully. (CS2 2026)
Solution: Using (1.1.34.1) for `a=*(&b)`:

Statement	a	b	c	ptr
a=5; b=11; c=20;	5	11	20	–
ptr=&a; *ptr=c;	20	11	20	&a
ptr=&c;	20	11	20	&c
a=*(&b);	11	11	20	&c
c=*ptr-a;	11	11	9	&c

TABLE 1.2.23: Trace

The output is 9.
The following code verifies the solution.

codes/q60.c

1.2.24 Consider the following ANSI-C function.

```c
int func(int start, int end){
    int length=end+1-start;
    if((length<1)||(start<0)||(end<0)){ return(0); }
    if(length%3==0){
        return(func(start+1, end));
    } else if(length%3==1){
        return(1+func(start, end-1));
    } else {
        return(func(start+2, end));
    }
}
```

The maximum possible value that can be returned from this function is _____. (*answer in integer*)
Note: Ignore syntax errors (if any) in the function. (CS2 2026)
Solution: Let $L = \texttt{end} + 1 - \texttt{start}$. The three cases give

$$L \equiv 0 \pmod 3 \implies L \to L - 1 \equiv 2, \text{ adds } 0 \tag{1.2.24.1}$$
$$L \equiv 1 \pmod 3 \implies L \to L - 1 \equiv 0, \text{ adds } 1 \tag{1.2.24.2}$$
$$L \equiv 2 \pmod 3 \implies L \to L - 2 \equiv 0, \text{ adds } 0 \tag{1.2.24.3}$$

Once $L \equiv 0$ or 2, it never becomes $\equiv 1$ again, and the recursion ends with 0. So 1 can be added at most once, for example

$$\texttt{func(0, 0)} = 1 + \texttt{func(0, -1)} = 1 \tag{1.2.24.4}$$

The maximum is 1.
The following code searches all `start`, `end` in $[-5, 299]$ for the maximum.

codes/func.c

1.2.25 The determinant of a 4×4 matrix A is 3. The value of the determinant of 2A is _____. (*answer in integer*) (CS2 2026)
Solution: From (1.1.23.1),

$$\det(2\mathbf{A}) = 2^4 \times 3 = 48 \tag{1.2.25.1}$$

The following code verifies the solution with a random matrix.

codes/det.py

1.2.26 Suppose an unbiased coin is tossed 6 times. Each coin toss is independent of all previous coin tosses. Let E_1 be the event that among the second, fourth, and sixth coin tosses, there are at least two heads. Let E_2 be the event that among the first, second, third, and fifth coin tosses, there are equal number of heads and tails. The conditional probability $\Pr(E_1 \mid E_2)$ is equal to _____. (*rounded off to one decimal place*) (CS2 2026)
Solution: Let $X_i = 1$ if toss i is a head and 0 otherwise, with $\Pr(X_i = 1) = \Pr(X_i = 0) = \frac{1}{2}$. Define

$$S_1 = X_2 + X_4 + X_6 \qquad \text{(heads among tosses 2, 4, 6)} \tag{1.2.26.1}$$
$$S_2 = X_1 + X_2 + X_3 + X_5 \qquad \text{(heads among tosses 1, 2, 3, 5)} \tag{1.2.26.2}$$

Then $E_1 = \{S_1 \geq 2\}$ and $E_2 = \{S_2 = 2\}$. From (1.1.9.1),

$$\Pr(E_1 \mid E_2) = \frac{\Pr(S_1 \geq 2,\ S_2 = 2)}{\Pr(S_2 = 2)} \tag{1.2.26.3}$$

Joint Z-transform. Each toss adds to S_1, S_2 or both:

Toss	Adds to	M_{X_i} in (z_1, z_2)
2	S_1 and S_2	$\frac{1}{2} + \frac{1}{2}z_1^{-1}z_2^{-1}$
4, 6	S_1 only	$\frac{1}{2} + \frac{1}{2}z_1^{-1}$
1, 3, 5	S_2 only	$\frac{1}{2} + \frac{1}{2}z_2^{-1}$

TABLE 1.2.26: Contribution of each toss

using (1.1.3.1) for each toss. The tosses are independent, so from (1.1.5.2),

$$M_{S_1,S_2}(z_1, z_2) = \left(\frac{1 + z_1^{-1}z_2^{-1}}{2}\right)\left(\frac{1 + z_1^{-1}}{2}\right)^2\left(\frac{1 + z_2^{-1}}{2}\right)^3 \tag{1.2.26.4}$$

Extracting $S_2 = 2$. From (1.1.5.1), the terms of (1.2.26.4) containing z_2^{-2} give $\Pr(S_1 = k, S_2 = 2)$. Expanding

$$\left(\frac{1 + z_2^{-1}}{2}\right)^3 = \frac{1}{8}\left(1 + 3z_2^{-1} + 3z_2^{-2} + z_2^{-3}\right) \tag{1.2.26.5}$$

a factor z_2^{-2} arises in two ways:

- toss 2 is a tail (the $\frac{1}{2}$ term) and two of tosses 1, 3, 5 are heads (the $\frac{3}{8}z_2^{-2}$ term);
- toss 2 is a head (the $\frac{1}{2}z_1^{-1}z_2^{-1}$ term) and one of tosses 1, 3, 5 is a head (the $\frac{3}{8}z_2^{-1}$ term).

So the coefficient of z_2^{-2} is

$$G(z_1) = \left(\frac{1}{2}\cdot\frac{3}{8} + \frac{1}{2}\cdot\frac{3}{8}z_1^{-1}\right)\left(\frac{1 + z_1^{-1}}{2}\right)^2 = \frac{3}{64}\left(1 + z_1^{-1}\right)^3 \tag{1.2.26.6}$$

$$= \frac{3}{64}\left(1 + 3z_1^{-1} + 3z_1^{-2} + z_1^{-3}\right) = \sum_{k=0}^{3}\Pr(S_1 = k, S_2 = 2)\, z_1^{-k} \tag{1.2.26.7}$$

Result. Setting $z_1 = 1$ in (1.2.26.7) adds all the coefficients:

$$\Pr(S_2 = 2) = G(1) = \frac{3}{64}\cdot 8 = \frac{3}{8} \tag{1.2.26.8}$$

and the coefficients of z_1^{-2} and z_1^{-3} give

$$\Pr(S_1 \geq 2,\ S_2 = 2) = \frac{3}{64}(3 + 1) = \frac{3}{16} \tag{1.2.26.9}$$

Substituting in (1.2.26.3),

$$\Pr(E_1 \mid E_2) = \frac{3/16}{3/8} = \frac{1}{2} = 0.5 \tag{1.2.26.10}$$

Note that $\frac{G(z_1)}{G(1)} = \left(\frac{1+z_1^{-1}}{2}\right)^3$: given E_2, S_1 has the binomial$\left(3, \frac{1}{2}\right)$ PMF of (1.1.4.1), as if toss 2 were still a fair coin.

The following code computes the joint Z-transform (1.2.26.4) as a coefficient array,

extracts $G(z_1)$, and checks the result by simulation in Fig. 1.2.26.1, where $K = S_1$ given E_2.

codes/coins.py

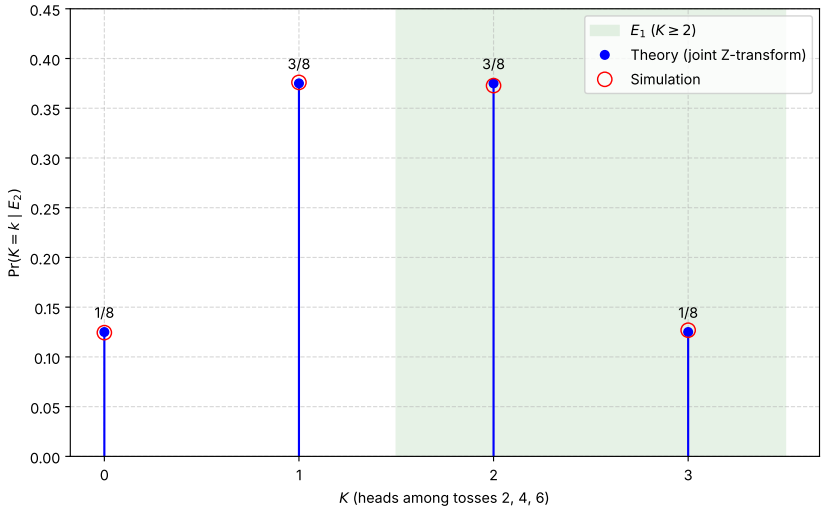


Fig. 1.2.26.1: PMF of K given E_2

1.2.27 Consider a function $f : (0,1) \to \{0,1\}$ defined as follows. For a real number $r \in (0,1)$, $f(r) = 1$ if the second digit after the decimal point in r is one of the four digits 2, 3, 6 and 7. Otherwise, $f(r)$ is equal to 0. The number of points in $(0,1)$ at which f is discontinuous is _____. (*answer in integer*) (CS2 2026)

Solution: From (1.1.35.1), f is constant on each $\left[\frac{k}{100}, \frac{k+1}{100}\right)$, so it can only jump at $r = \frac{k}{100}$, $k = 1, \ldots, 99$, where the digit changes from $d-1$ to $d = k \bmod 10$ (from 9 to 0 when $d = 0$).

d	0	1	2	3	4	5	6	7	8	9
f	0	0	1	1	0	0	1	1	0	0
Jump	No	No	Yes	No	Yes	No	Yes	No	Yes	No

TABLE 1.2.27: Jumps of f

f jumps when $k \bmod 10 \in \{2,4,6,8\}$, which happens $4 \times 10 = 40$ times for $k = 1, \ldots, 99$. So f has 40 points of discontinuity.
The following code counts the jumps and plots f in Fig. 1.2.27.1.

codes/digit.py

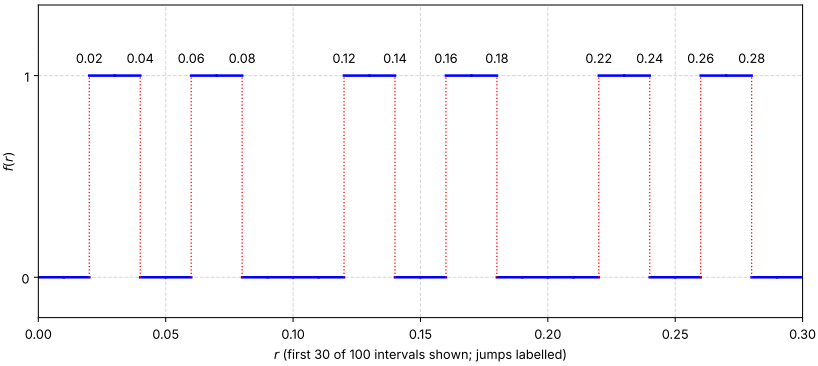


Fig. 1.2.27.1: $f(r)$ near the start of $(0, 1)$